5. Isomorphism

Consider diagrams G_1 and G_2.

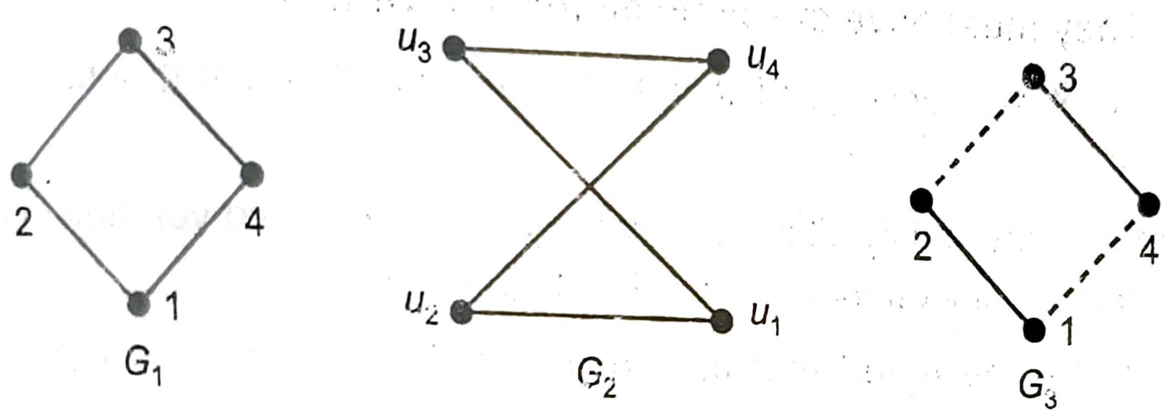


Fig. 7.15

Do they represent the same graph ? If we change the names of vertices in G_2 as $u_1 \rightarrow 2$, $u_2 \rightarrow 1$, $u_3 \rightarrow 3$, $u_4 = 4$, then we see that the second diagram can be redrawn to get the first diagram. Imagine that you "lift" the vertex u, and put it on the left. 'Essentially' the two graphs are the 'same' although they 'look' different. Such graphs are called **isomorphic**, (iso = same, morph = form).

Definition : Two graphs $G = (V, E)$ and $G' = (V', E')$ are said to be **isomorphic**, if there exists a one-to-one correspondence f from V to V' such that if $v_1, v_2 \in V$ and $v_1', v_2' \in V'$ and $f(v_1) = v_1'$, $f(v_2) = v_2'$ then the number of edges between v_1, v_2 is the same as the number of edges between v_1' and v_2'. The function f is called an **isomorphism** between G and G'. If G and G' are isomorphic we write $G \cong G'$.

Note that

(i) If f is an isomorphism between G and G' the f^{-1} is also an isomorphism.

(ii) If G is isomorphic to G' and G' is Isomorphic to G'' then G is isomorphic to G''.

(iii) If $v_1, v_2 \in V$ are adjacent then $v_1', v_2' \in V'$ are also adjacent.

(iv) If $e = \{v_1, v_1\}$ is a loop, then $e' = \{v_1', v_1'\}$ is also a loop.

(v) If $e_1 = \{v_1, v_2\}$ and $e_2 = \{v_2, v_3\}$ are two adjacent edges *i.e.* e_1, e_2 are two edges incident at a common vertex v_2 then $e_1' = \{v_1', v_2'\}$ and $e_2' = \{v_2', v_3'\}$ are also adjacent edges.

(vi) If $e_1 = \{v_1, v_2\}$, $e_2 = \{v_1, v_2\}$ are parallel edges then $e_1' = \{v_1', v_2'\}$, $e_2' = \{v_1', v_2'\}$ are also parallel edges. In other words if G is a multigraph then G' is also a multigraph.

(vii) $d(v) = d(v')$ for all $v \in V$ *i.e.* degree of vertex v is equal to the degree of vertex v' under f.

It is not always easy to determine whether two graphs are isomorphic.

If two graphs are isomorphic then,

1. **They must have the same number of vertices,**
2. **They must have the same number of edges.**
3. **They must have the same degrees of vertices.**

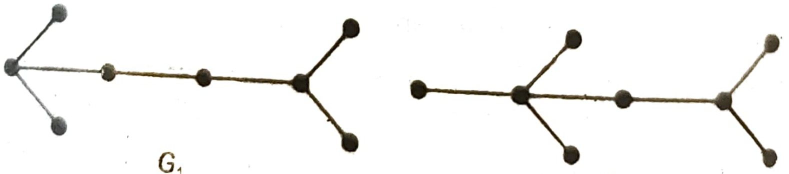
If one of the above conditions is not satisfied then the graphs are not isomorphic.

Graphs may satisfy all the above three conditions and yet they may not be isomorphic. For example, see the Ex. 1 (a) below.

4. It may be noted that in addition to the above three conditions 'adjacency' also is an important condition.

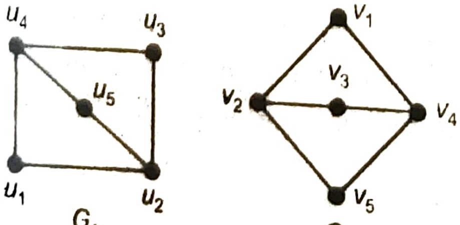
Example 1 : Determine whether the following pairs of graph are isomorphic.

(a)



G_1 G_2

(b)



G_1 G_2

Sol. : (a) The two graphs G_1, G_2 have eight vertices and seven edges. However, in the first graph G_1, there are four vertices of degree one and in the second graph G_2, there are five vertices of degree one.

Hence, the two graphs are **not** isomorphic.

(b) The two graphs G_1, G_2 have five vertices and six edges each. Further, in both the graphs there are three vertices of degree two and two vertices of degree three.

We can define one-to-one correspondence f as follows :

$$u_1 \to v_5,\ u_2 \to v_4,\ u_3 \to v_1,\ u_4 \to v_2,\ u_5 \to v_3.$$

This correspondence preserves the adjacency and incidence relationship.

$\therefore$ The graphs are isomorphic.

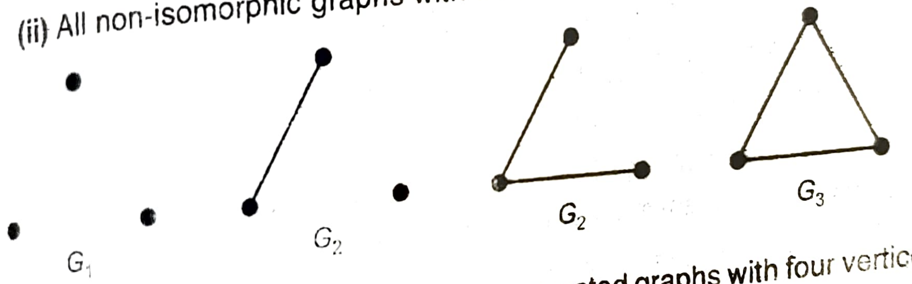
Note ....

G can be obtained by just terning G_1 through 125°.

Example 2 : Draw all non-isomorphic graphs of (i) 2 vertices, (ii) 3 vertices and state reasons.

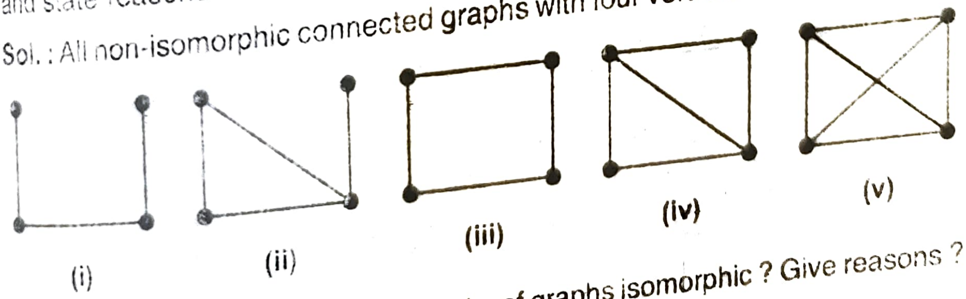
Sol. : (i) All non-isomorphic graphs with two vertices are

(ii) All non-isomorphic graphs with three vertices are



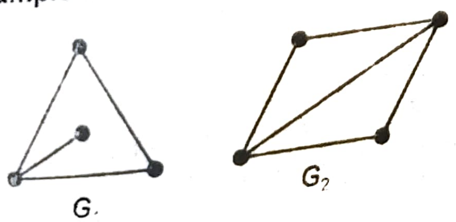
Example 3 : Find all non-isomorphic connected graphs with four vertices and state reasons.

Sol. : All non-isomorphic connected graphs with four vertices are shown below.



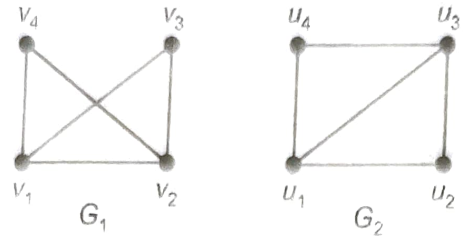
Example 4 : Are the following pairs of graphs isomorphic ? Give reasons ?

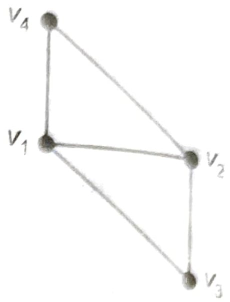
(i)



Sol. : No. Although G_1 and G_2 have 4 vertices both: G_1 has 4 edges and G_2 has 5 edges; G_1 has a pendant vertex, G_2 has not. G_1 has only one vertex with 3 edges while G_2 has two vertices with 3 edges.

(ii)

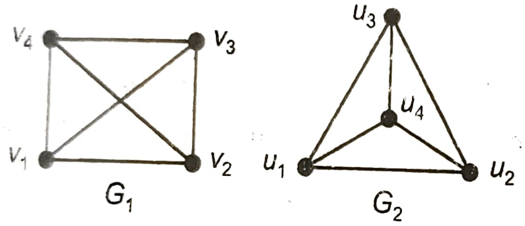


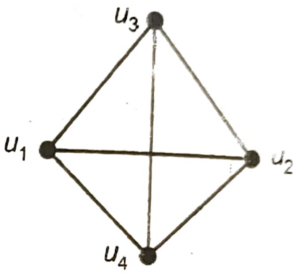


Sol. : Yes. G_1 and G_2 both have 4 vertices; 5 edges; 2 vertices of degree 2 and 2 vertices of degree 3. The correspondence is obtained by **lifting** v_3 and placing it below v_2.

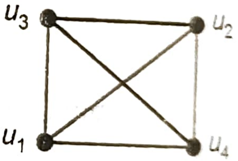
$\therefore\ v_1 \to u_1,\ v_3 \to u_2,\ v_2 \to u_3,\ v_4 \to u.$

(iii)



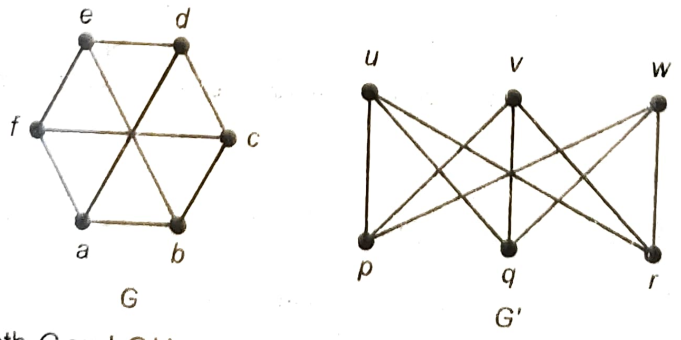


Sol. : Yes. G_1 and G_2 both have 4 vertices 4 edges. Each vertex in G_1 and G_2 is of degree 3. The correspondence is obtained if u_4 is pulled down and the figure is rotated through 45°.



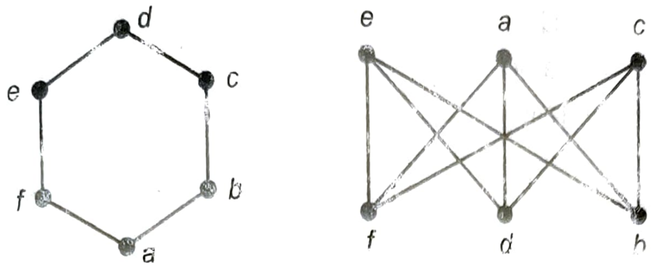
$\therefore\ v_1 \to u_1,\ v_2 \to u_4,\ v_3 \to u_2,\ v_4 \to u_3.$

Example 5 : Are the following graphs isomorphic ? Give reasons.



Sol. : Yes. Both G and G' have 6 vertices and 9 edges. Each vertex in G and G' is of degree 3. The correspondence is obtained if G is rotated through 45° (in anticlockwise direction) and d and a are interchanged by pushing a up and pulling d down.

$\therefore\ e \to u,\ a \to v,\ c \to w,\ f \to p,\ d \to q,\ b \to r.$

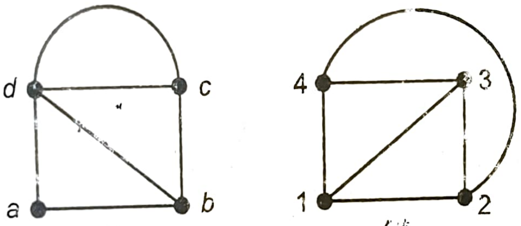


Example 6 : Find whether the following graphs $G = (V, E)$ and $G^* = (V^*, E^*)$ are isomorphic giving reasons.

(i) $V = \{a, b, c, d\}$, $E = \{(a, b), (a, d), (b, d), (c, d), (c, b), (c, d)\}$

(ii) $V^* = \{1, 2, 3, 4\}$, $E^* = \{(1, 2), (2, 3), (3, 1), (3, 4), (4, 1), (4, 2)\}$

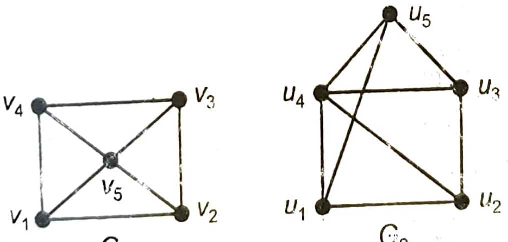
Sol. : (i) We shall first draw the diagrams of these graphs which are given in set notation. In E the edge (c, d) occurs twice, indicating that there are two edges between c and d.



G G^*

(ii) No. Both graphs G and G^* have 4 vertices and 6 edges. But the degree of vertex a in G is 2 and there is no vertex of degree 2 in G^*. Also in G the degree of vertex d is 4, but there is no vertex of degree 4 in G.

Example 7 : Find whether the follows graphs are isomorphic.



G_1 G_2

(M.U. 2000)

Sol. : We first note that both the graphs have

1. five vertices
2. eight edges

In G_1 there is one vertex v_5 of degree 4 and the remaining vertices are of degree 3.

In G_2, there is one vertex u_4 of degree 4 and the remaining are of degree 3.

Further, the adjancy property is observed. The vertex with degree 4 is adjacent to the remaining vertices in both the graphs.

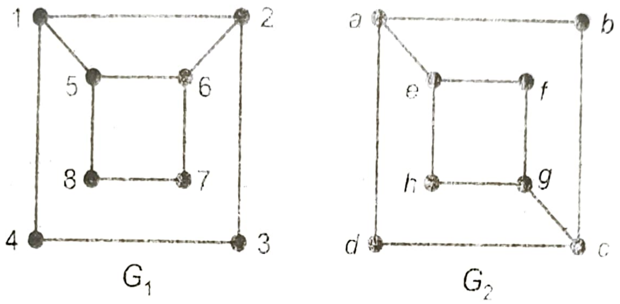
Hence, the two graphs are isomorphic.

If you pull u_4 inside and u_5 to the position of u_4, you will get G_1.

The correspondence is

$$v_1 \to u_1,\ v_2 \to u_2,\ v_3 \to u_3,\ v_4 \to u_5 \text{ and } v_5 \to u_4.$$

Example 8 : Determine whether the following graphs are isomorphic.



(M.U. 2002)

Sol. : There are 8 vertices in G_1 and 8 vertices in G_2.

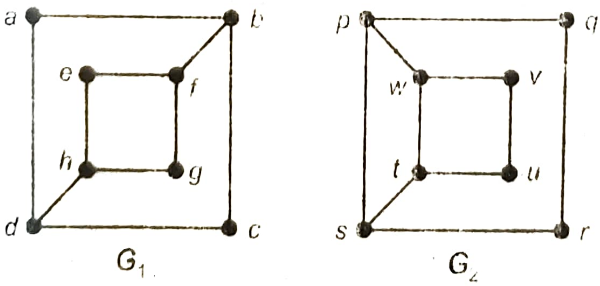
Further there are four vertices of degree 3 in G_1 and four vertices of degree 3 in G_2. There are four vertices of degree 2 in G_1 and there are four vertices of degree 2 in G_2.

But if $5 \to e$ and $6 \to g$ then although 5 and 6 are adjacent, e and g are not adjacent [See (4), 7-9]

Similarly, if $1 \to a$ and $2 \to c$ although 1 and 2 are adjacent but a and c are not adjacent. Hence, the graphs are not isomorphic. Thus, we see that the **adjacency** is not preserved.

The graphs are **not** isomorphic.

Example 9 : Determine whether the graphs G_1 and G_2 are isomorphic or not. Justify your answer.



Sol. : We first note the following.

1. Both the graphs have the same number of vertices *viz.* 8 and the same number of edges 10.

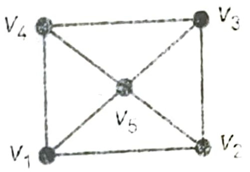
2. In G_1 there are four vertices with degree 3 and in G_2 also there are four vertices with degree 3.

3. But adjancy is not preserved in the two graphs. In G_1 vertex with 3 edges is adjacent to only one vertex with 3 edges (f and b or d and h). But in G_2 a vertex with 3 edges in adjacent to two vertices with edges 3 (p to w and s; w to p and t and so on).

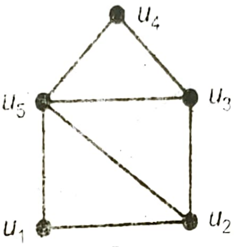
4. Hence G_1 and G_2 as above are **not** isomorphic.

Example 10 : Show that the following graphs are isomorphic.

(M.U. 1998, 2000)



G_1



G_2

Sol. : We first see that the two graphs have the same number of vertices (5) and the same number of edges (8).

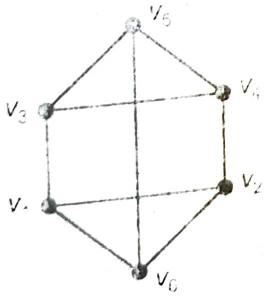
In G_1 there are 4 vertices of degree 3 and one vertex of degree 4. In G_2 also there are 4 vertices of degree 3 and one vertex of degree 4.

Consider the correspondence $v_1 \to u_1$, $v_2 \to u_2$, $v_3 \to u_3$, $v_4 \to u_4$ and $v_5 \to u_5$.

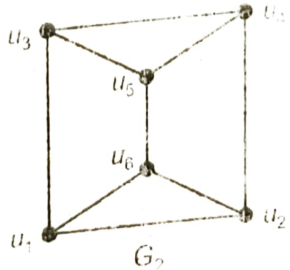
(If we pull u_5 towards u_2, so that u_4 becomes the vertex of the rectangle G_2 will look like G_1).

Hence, G_1 and G_2 are isomorphic.

Example 11 : Determine whether the following graphs are isomorphic or not.



G_1



G_2

Sol. : We first see that both the graphs have the same number of vertices (6) and the same number of edges (9).

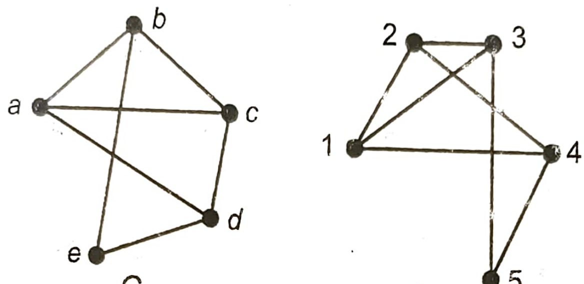
In G_1 all the vertices are of degree 3 and in G_2 also all the vertices are of degree 3.

Consider the correspondence $v_1 \to u_1$, $v_2 \to u_2$, $v_3 \to u_3$, $v_4 \to u_4$, $v_5 \to u_5$, $v_6 \to u_6$.

(If we press the edge (v_5, v_6) in G_1 and bring it within the rectangle after making it short, we will get the graph G_2.)

Hence, G_1 and G_2 are isomorphic.

Example 12 : Show that the following two graphs are isomorphic.



G_1 G_2

Sol. : We first see that the two graphs have the same number of vertices (5) and the same number of edges (7).

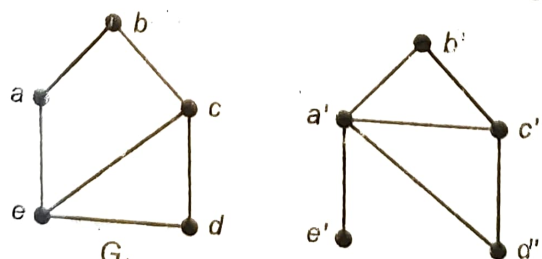
In G_1 there are 4 vertices of degree 3 and one vertex of degree 2. In G_2 also there are 4 vertices of degree 3 and one vertex of degree 2.

Consider the correspondence $a \to 1$, $b \to 2$, $c \to 3$, $d \to 4$, $e \to 5$.

(If we rotate G_2 keeping the vertex 1 fixed, in clockwise direction such that the edge (1, 3) becomes horizontal, then G_2 will look like G_1.)

Hence, G_1, G_2 are isomorphic.

Example 13 : Show that the following graphs are not isomorphic.



G_1 G_2

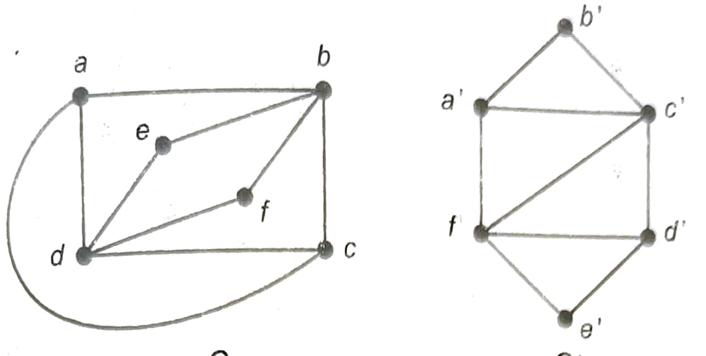
Sol. : Both the graphs have 5 vertices. Both the graphs have 6 edges.

But in graph G_2 there is one vertex a' with degree 4 and one vertex e' with degree 1.

There is no vertex with degree 4 and degree 1 in G_1.

$\therefore$ The graphs are not isomorphic.

Example 14 : Determine whether the following pair of graphs is isomorphic or not. **(M.U. 2012)**



G G'

Sol : We first note that both the graphs have

1. same number of vertices *viz.* 6
2. same number of edges *viz.* 9.

In G, there are two vertices e, f of degree 2, two vertices c and a of degree 3, two vertices b and d of degree 4.

In G', also there are two vertices b' and e' of degree 2, two vertices a' and d' of degree 3, two vertices c' and f' of degree 4.

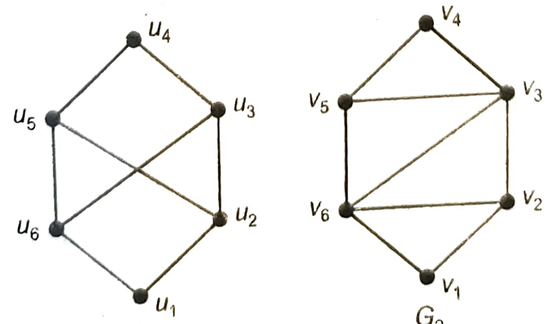
Yet the two graphs are **not** isomorphic because **the property of adjacency is not observed.**

In G, one vertex e (and also f) of degree 2 is adjacent to two vertices d and b of degree 4.

In G', one vertex b' (and also e') of degree 2 is adjacent to the vertex c' with degree 4 but to a' with degree 3.

(Try if you can to obtain G from G'.)

Example 15 : Determine whether the following graphs are isomorphic.



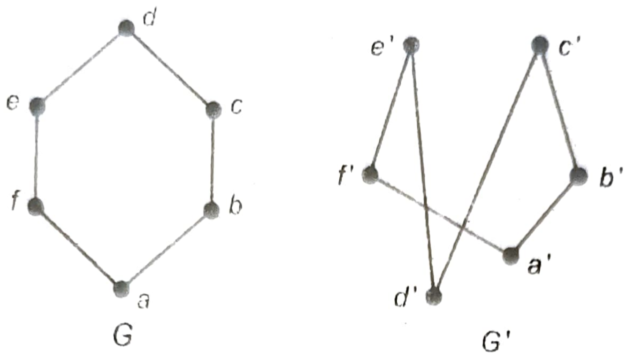
G_1 G_2

(M.U. 2003)

Sol. : We first see that both the graphs have the same number of vertices (6). But the graph G_1 has 8 edges and the graph G_2 has 9 edges.

Hence, the two graphs are not isomorphic.

Example 16 : Find whether the following graphs are isomorphic.



(M.U. 2000)

Sol. : We first note that both the graphs G and G' have

(i) the same number of vertices *viz.* 6

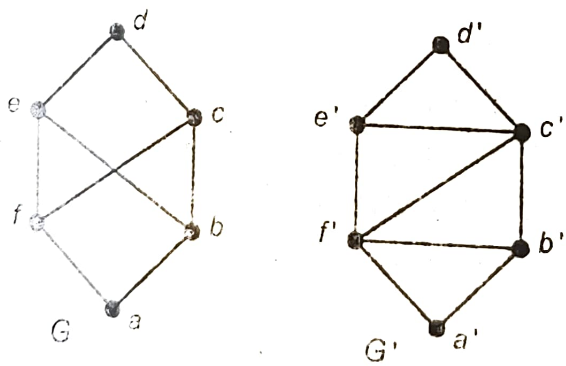
(ii) the same number of edges *viz.* 6

(iii) each vertex in G and G' is of degree 2.

Hence, the two graphs are isomorphic.

G can be obtained from G' by "pulling" the vertex d' up above the vertices c' and e'.

Example 17 : Determine whether the following graphs are isomorphic.



(M.U. 2010)

Sol. : We first note that G and G' have

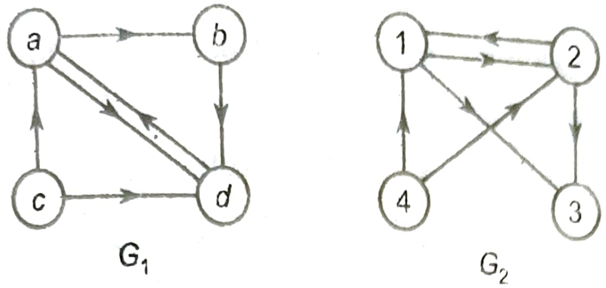
(i) the same number of vertices *viz.* 6

(ii) but they do not have the same number of edges. G has 8 edges while G' has 9 edges.

(iii) In G' the vertex c' is of degree 4, while in G there is no vertex of degree 4.

Hence, G and G' are **not** isomorphic.

Example 18 : Determine whether the following graphs are isomorphic.



(M.U. 2009)

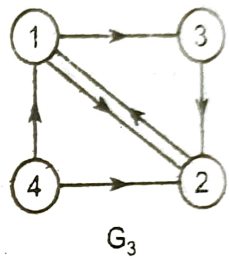
Sol. : We first observe that

(i) G_1 and G_2 have the same number of vertices, 4.

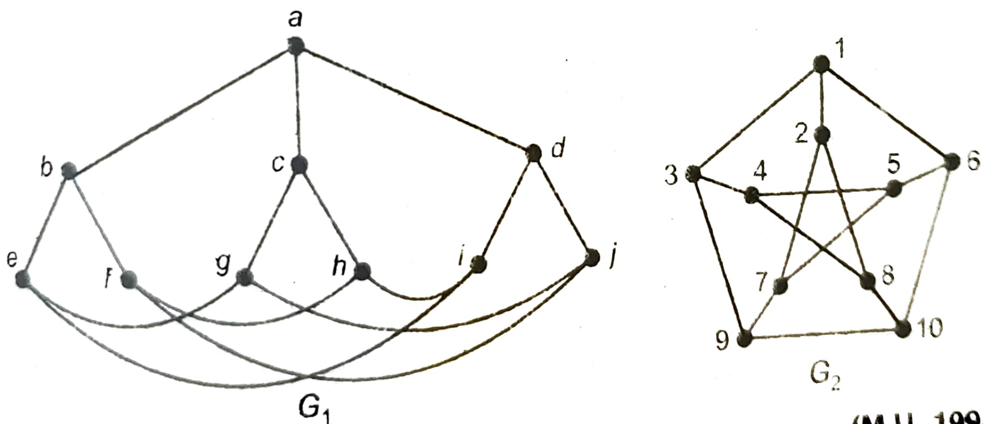
(ii) G_1 and G_2 have the same number of edges, 6.

(iii) Both G_1 and G_2 have two vertices of degree 4 and two vertices of degree 2.

If we interchange the positions of (2) and (3) in G_2, we get G_3 as shown in adjoining figure.



Example 19 : Discuss whether the following graphs are isomorphic.



(M.U. 1998)

Sol. : We first see that both the graphs have the same number of vertices (10) and the same number of edges (15).

In both the graphs all the 10 vertices are of degree 3.

Also the adjacency is preserved.

Consider the correspondence $a \to 1, b \to 2, c \to 3, d \to 4, e \to 5, f \to 6, g \to 7, g \to 8, i \to 9, j \to 10$.

Hence, the two graphs are isomorphic.